Manufacturing

MEE 101 Introduction to Mechanical Engineering

Lecture Slides #3

Manufacturing

- Once the detailed design has been completed, a designer will be involved with the manufacturing of the product.
- The fabrication techniques that an engineer selects will depend on the time and expense of setting up the tooling and machines necessary.
- Some systems—for instance, automobiles, air conditioners, microprocessors, hydraulic valves, and computer hard disk drives—are mass produced, a term that denotes the widespread use of mechanical automation.

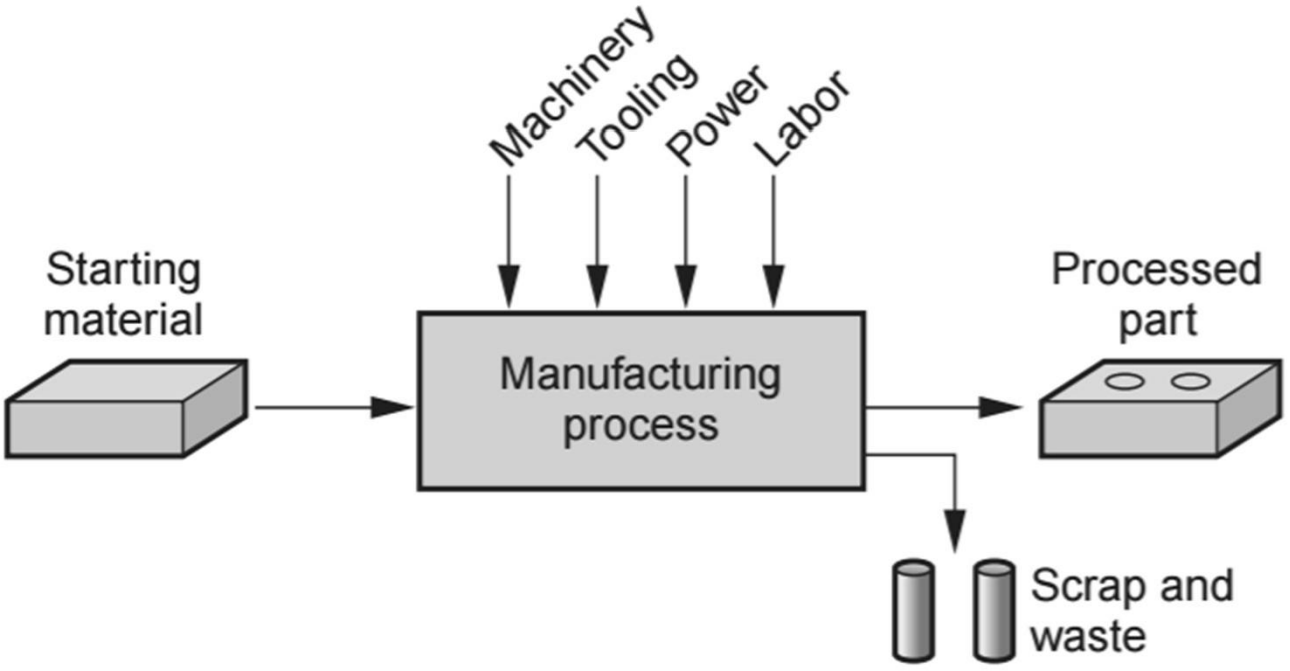
Manufacturing

The development of flexible manufacturing systems allows a production line to quickly reconfigure to different components for different vehicles.



Manufacturing Processes

- Manufacturing technologies are economically important because they are the means for adding value to raw materials by converting them into useful products.

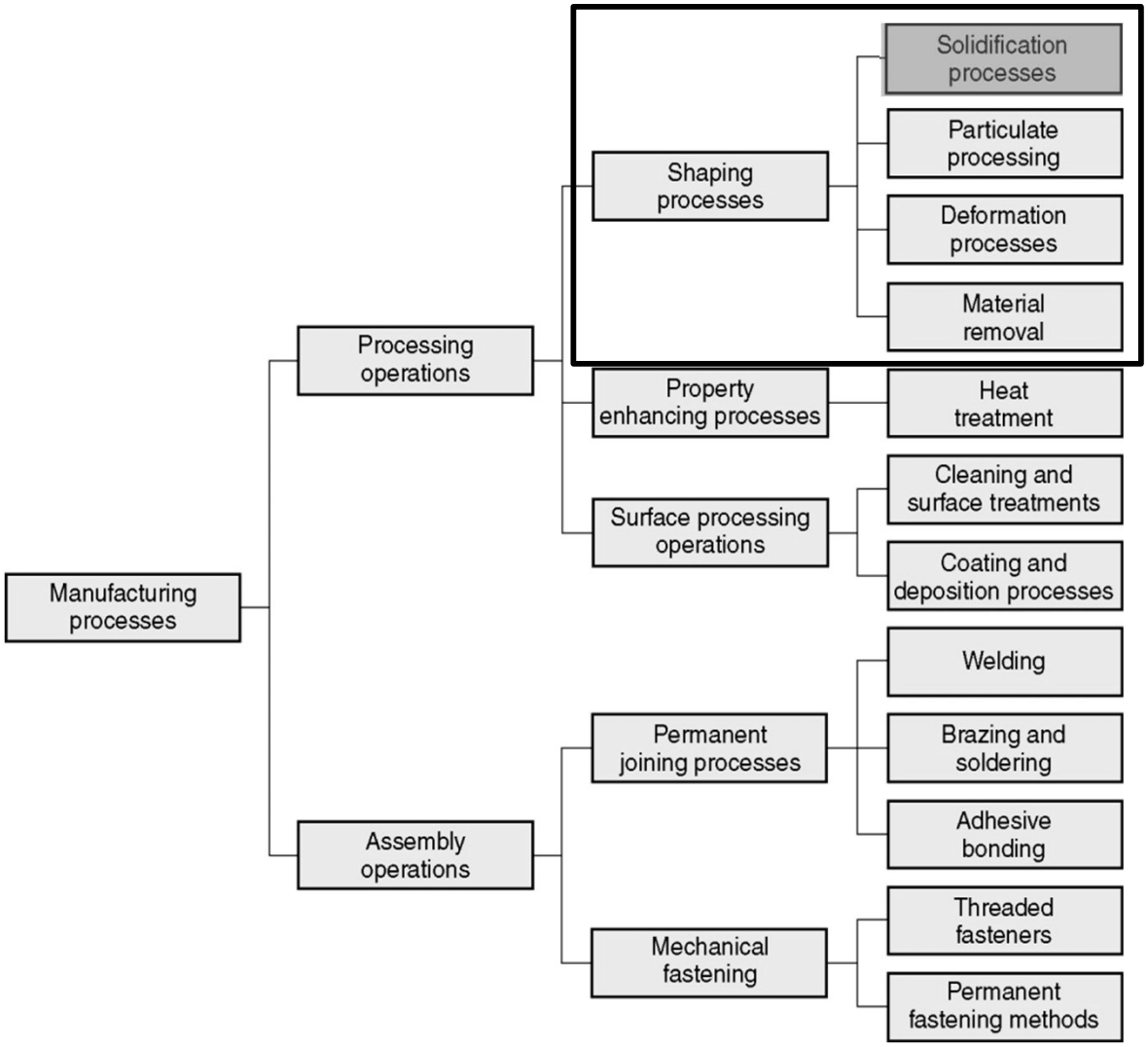


The Ingenious Design of the Aluminum Beverage Can

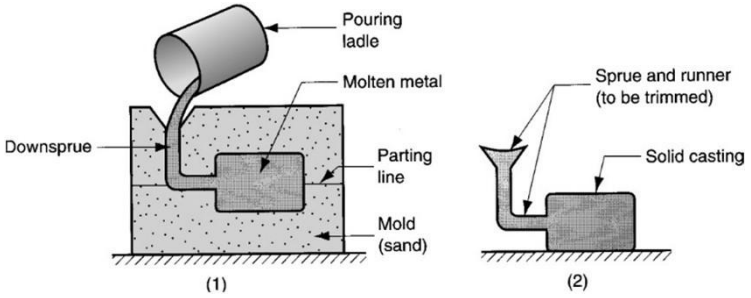
Manufacturing Processes

- There are many different manufacturing processes, and each is well suited to a particular need based on
  - dimensional accuracy,
  - material properties,
  - mechanical component's shape,
  - environmental impact.
- Engineers select processes, identify the machines and tools, and monitor production to ensure that the final product meets its specifications.

Manufacturing Processes

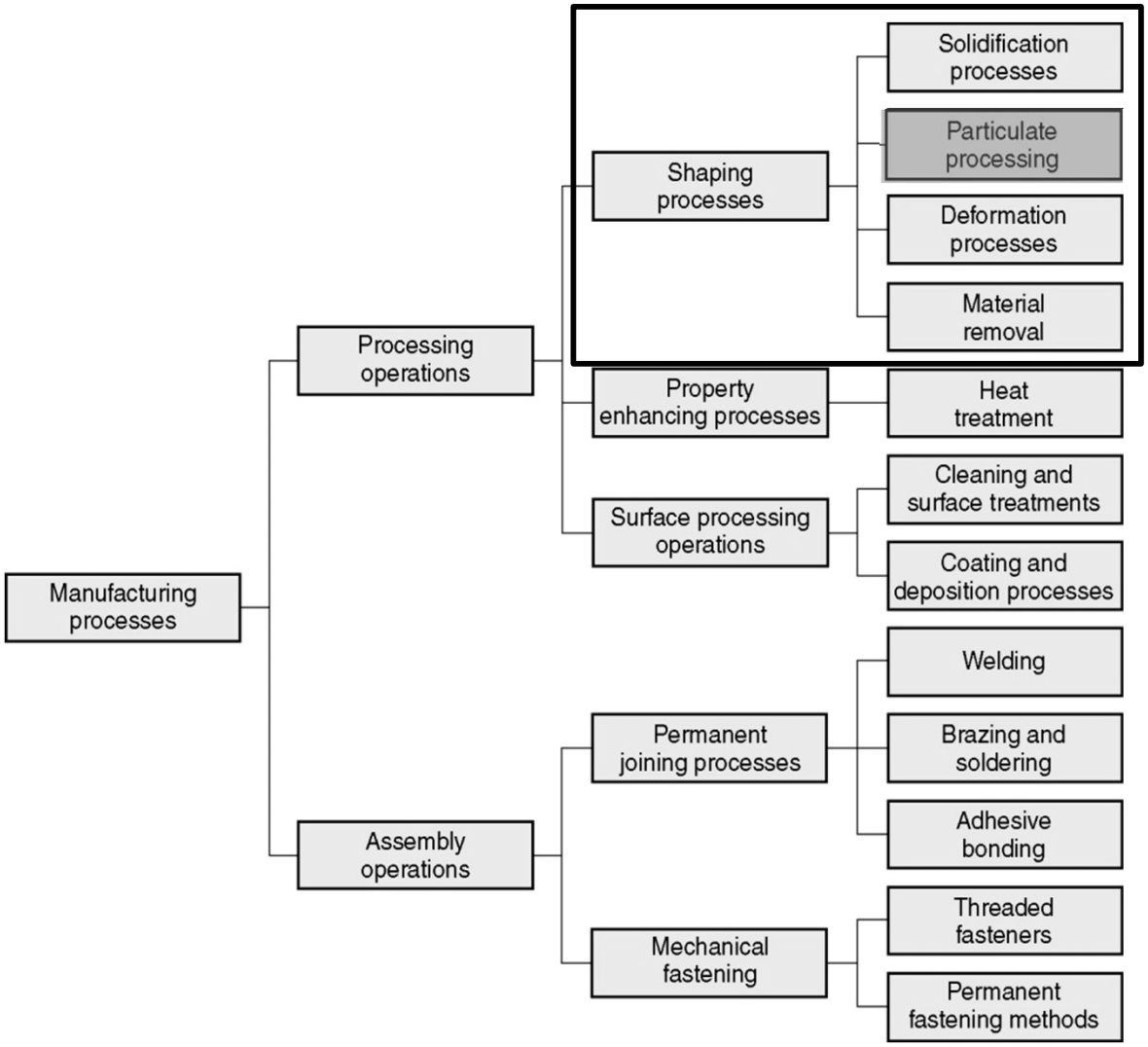


Solidification processes (Casting)

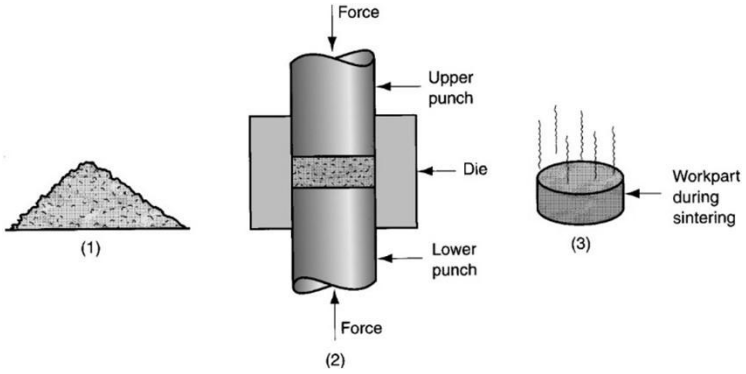


Starting material is heated sufficiently to transform it into a liquid or highly plastic state

Manufacturing Processes

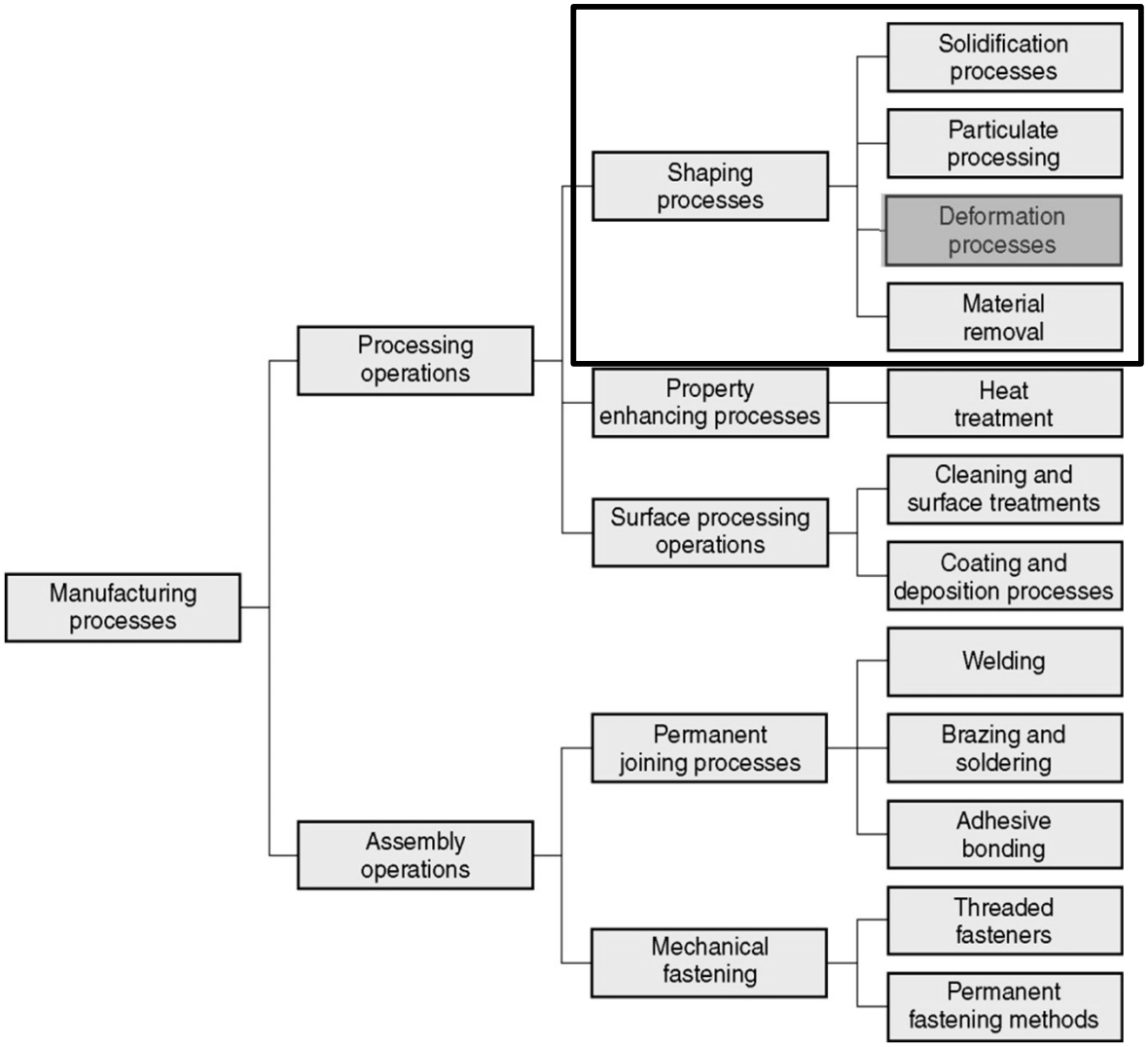


Particulate Processing (Powder metallurgy)

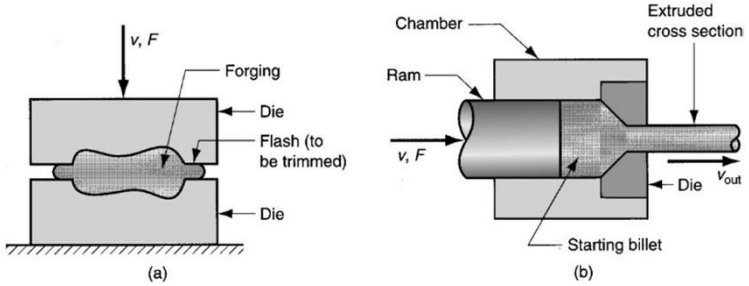


(1) Starting materials are metal or ceramic powders, which are (2) pressed and (3) sintered

Manufacturing Processes



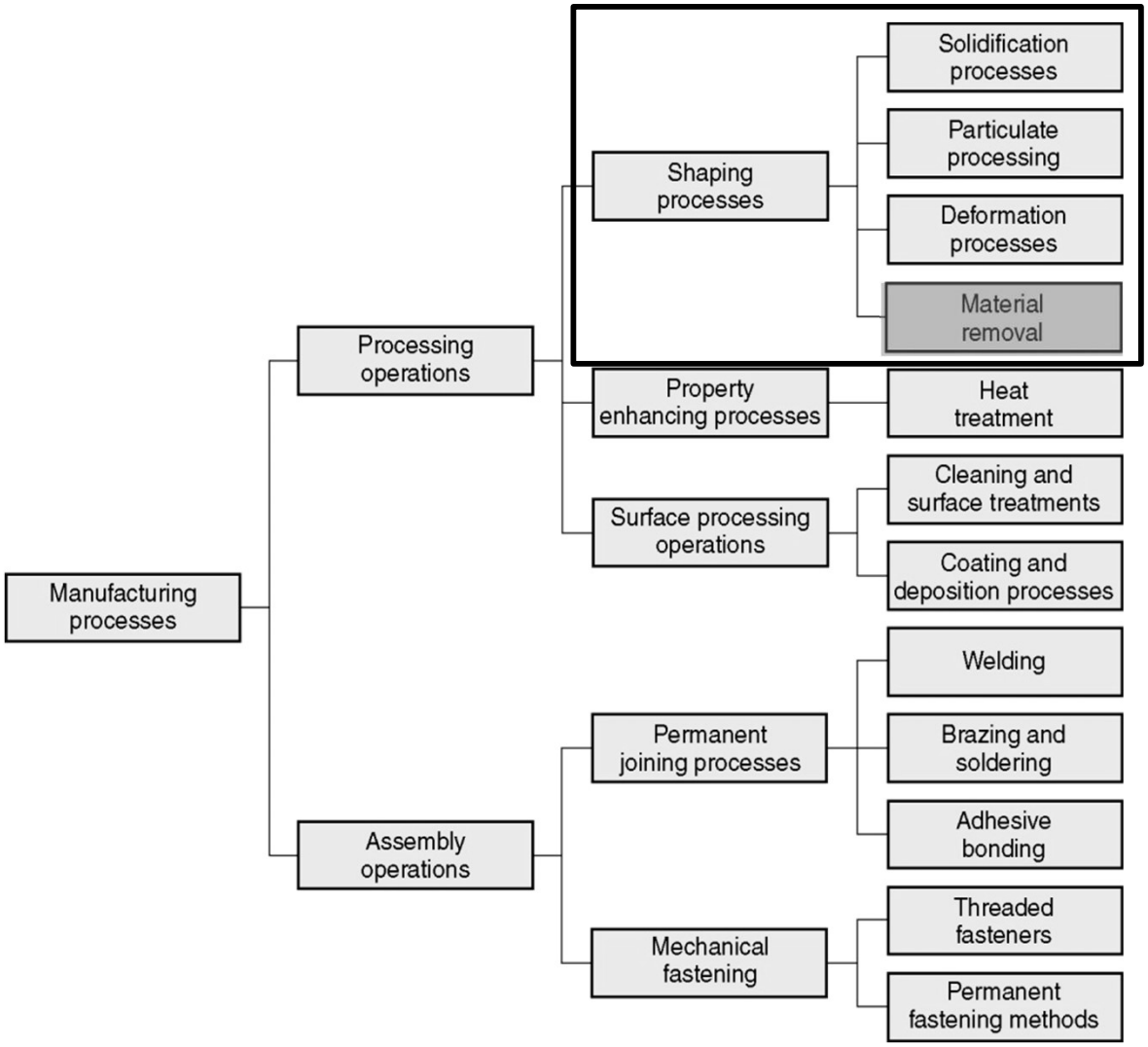
Deformation Processes (Forming)



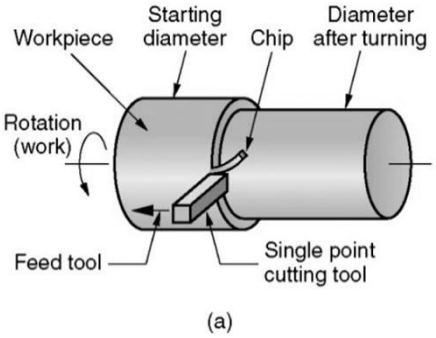
Starting workpart is shaped by application of forces that exceed the yield strength of the material

Examples: (a) forging and (b) extrusion

Manufacturing Processes



Material Removal Processes (Machining)



Excess material removed from the starting piece so what remains is the desired geometry

Examples: (a) turning, (b) drilling, and (c) milling

Manufacturing Processes: Casting

- Casting is the process in which liquid metal, such as gray iron, aluminum, or bronze, is poured into a mold, cooled, and solidified.
- Complex shapes can be produced as solid objects without the need to join any pieces together.
- Some examples of cast components include automotive engine blocks, cylinder heads, and brake rotors and drums.





Making the one ring to rule them all

Manufacturing Processes: Powder Metallurgy

Metal processing technology in which parts are produced from metallic powders
Usual PM production sequence:

- Pressing - powders are compressed into desired shape to produce green compact
- Accomplished in press using punch-and-die
- Sintering – green compacts are heated to bond the particles into a hard, rigid mass
- Temperatures are below melting point



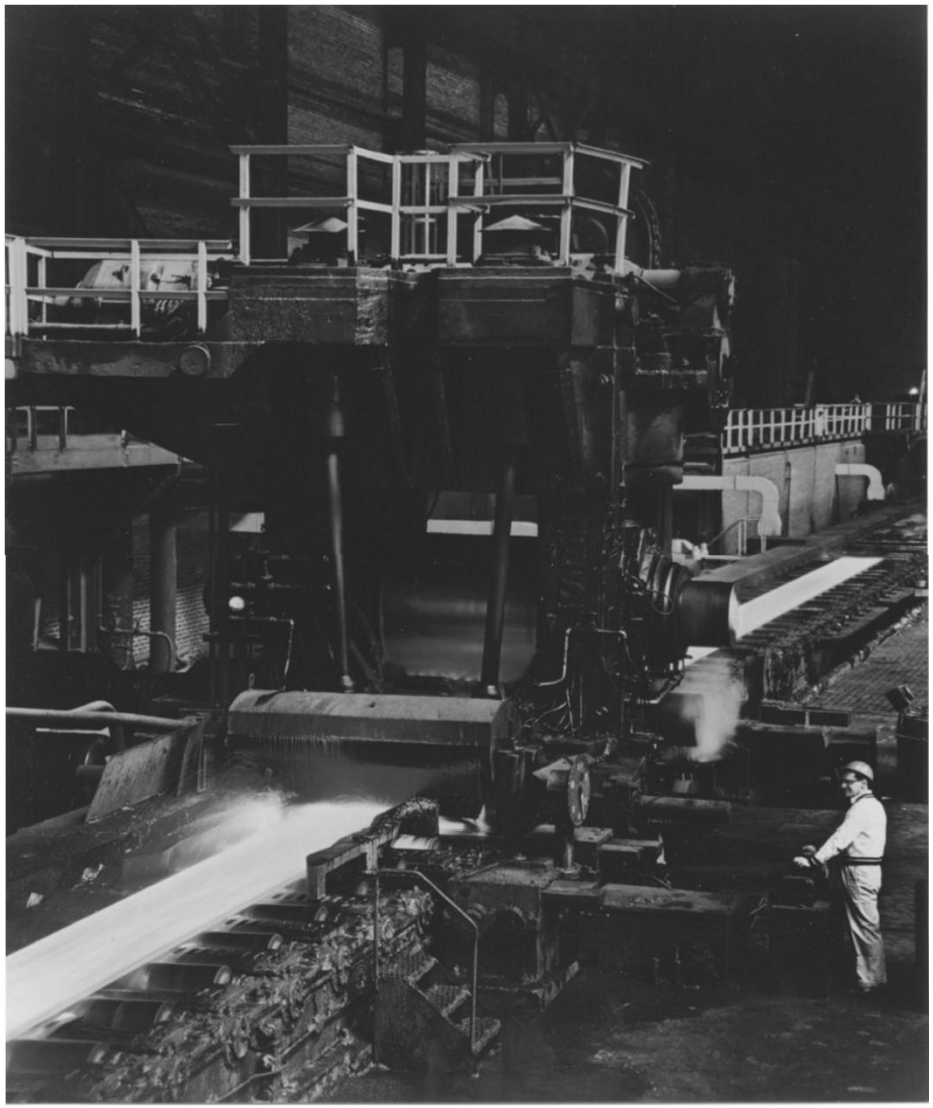
Powder metallurgy

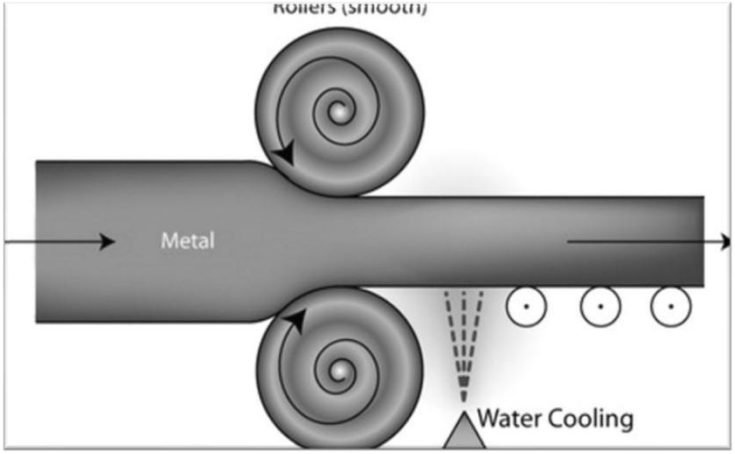
Manufacturing Processes: Forming

- Forming encompasses a family of techniques in which a raw material is shaped by stretching, bending, or compression.
- Large forces are applied to plastically deform a material into its new permanent shape.
- Most common forming techniques are *rolling*, *forging*, and *extrusion.*
- Other techniques include, *bending*, *cold forming*, *spinning*, etc.

Manufacturing Processes: Forming

- Rolling is the process of reducing the thickness of a flat sheet of material by compressing it between rollers.
- Rolling is classified according to the temperature of the metal rolled.
  - If the temperature of the material is above its recrystallization temperature, the process is called *hot rolling*,
  - If it is below recrystallization temperature, the process is *cold rolling*.

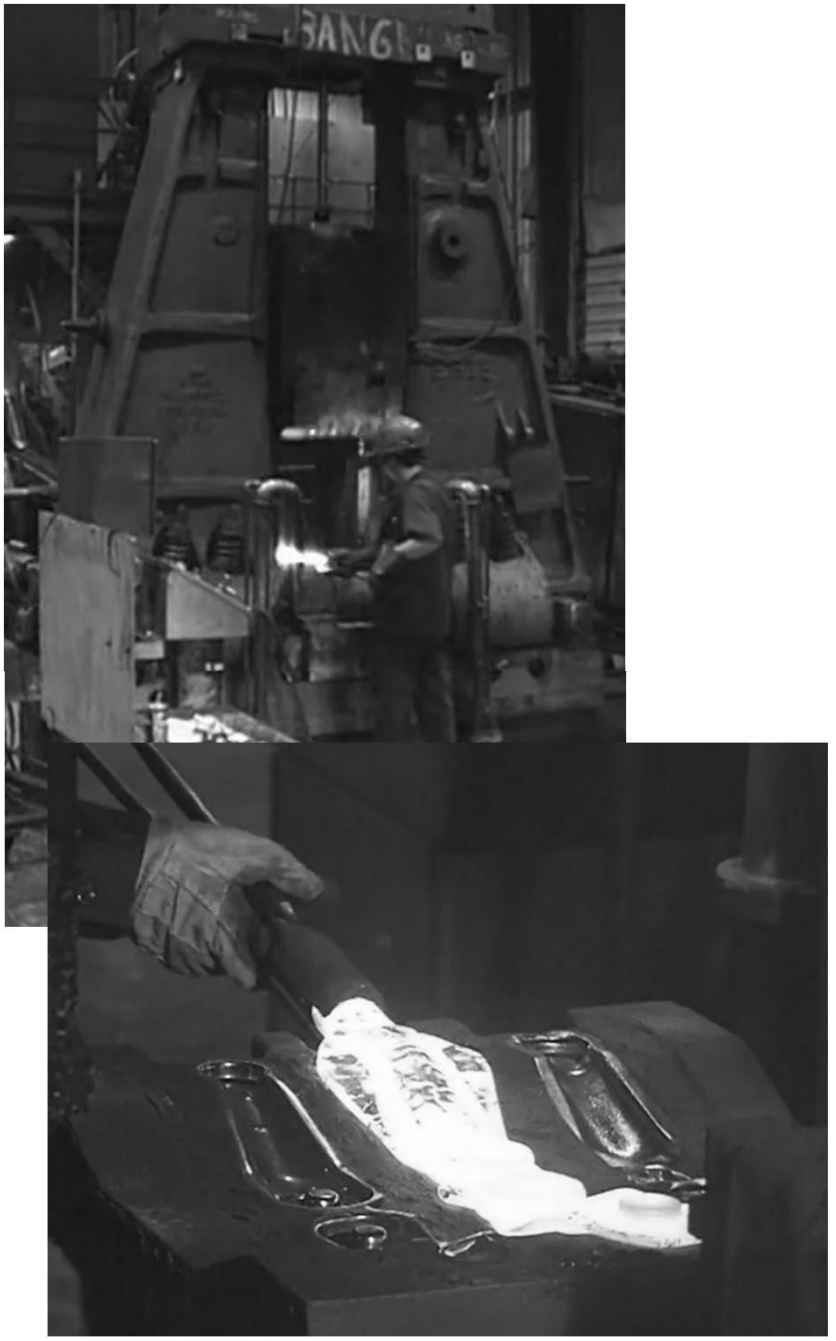


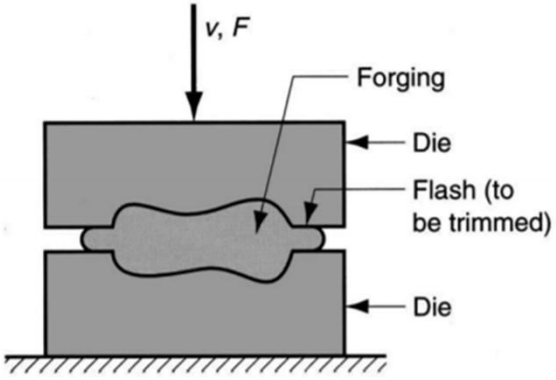


Steel Rolling

Manufacturing Processes: Forming

- Forging is another forming process involving the shaping of metal using localized compressive forces.
- It is based on the principle of heating, impacting, and plastically deforming metal into a final shape.
- The blows are delivered with a hammer or a die.

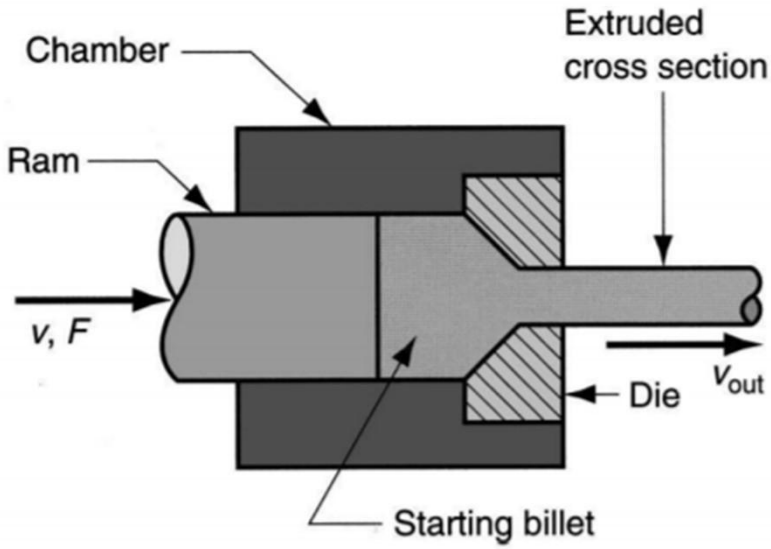


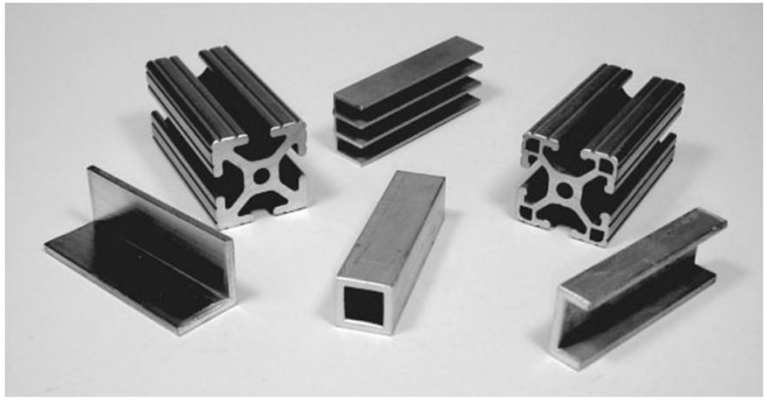


Forging

Manufacturing Processes: Forming

- Extrusion is used to create long straight metal/plastic parts with their cross sections having round, rectangular, L-, T-, or C-shapes, for instance.
- In extrusion, a mechanical or hydraulic press is used to force heated material through a tool (called a die) that has a tapered hole ending in the shape of the finished part's cross section.
- Conceptually, it is similar to squeezing toothpaste out of a tube.



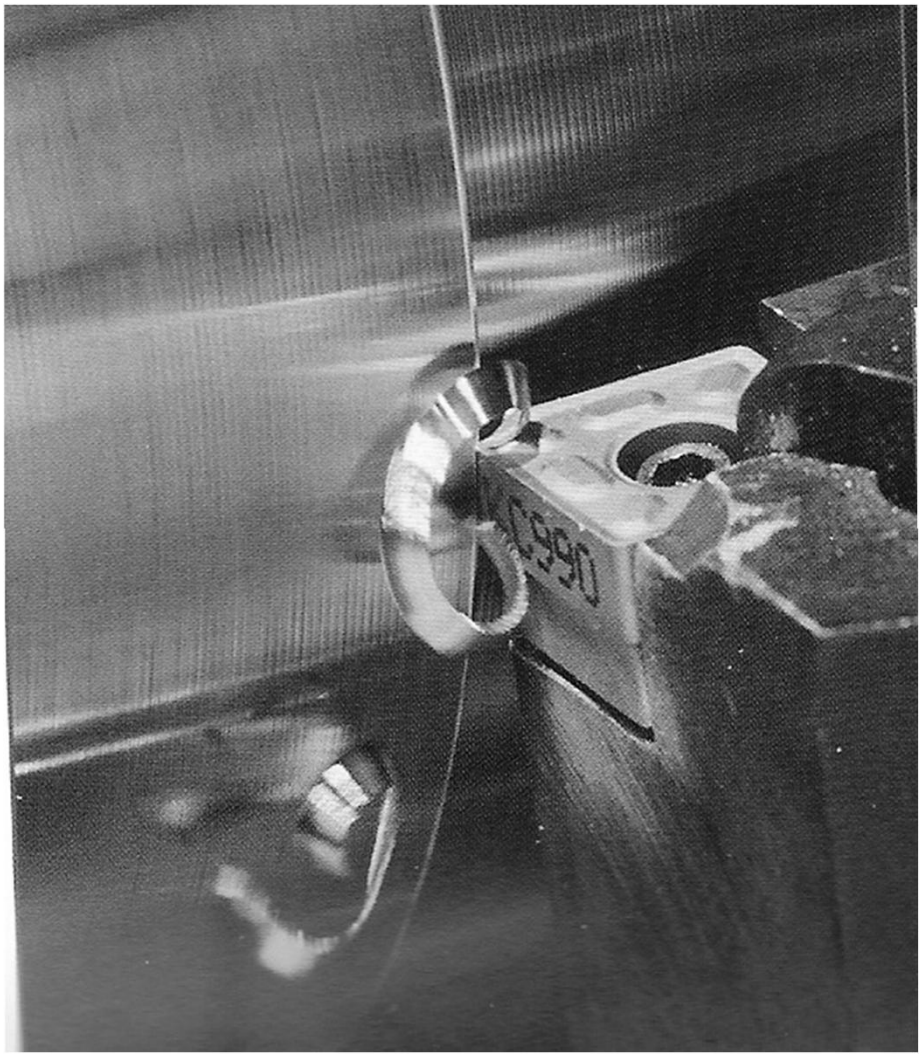


Extrusion process

Manufacturing Processes: Machining

- Machining refers to processes in which a sharp metal tool removes material in the form of small chips by cutting it.

- Machining operations are capable of producing mechanical components with dimensions and shapes that are far more precise than their cast or forged counterparts.

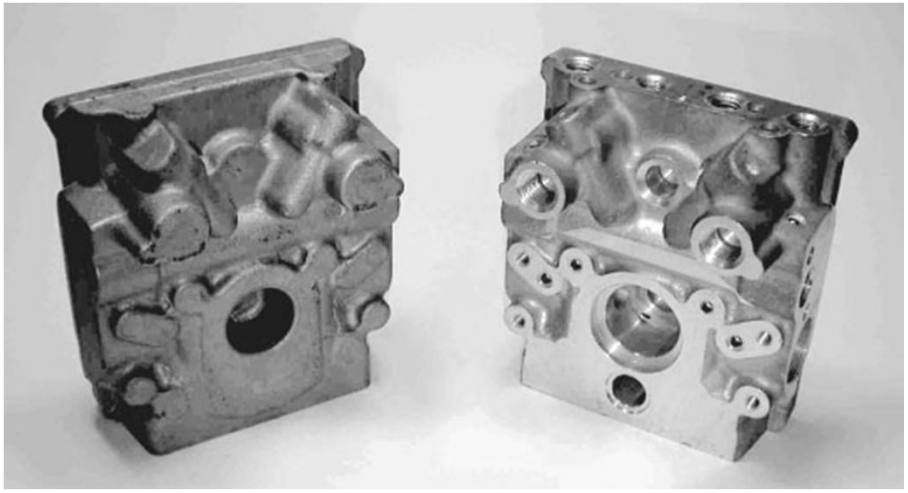


Chip formation in Machining

Chip Formation in Machining 2

Manufacturing Processes: Machining

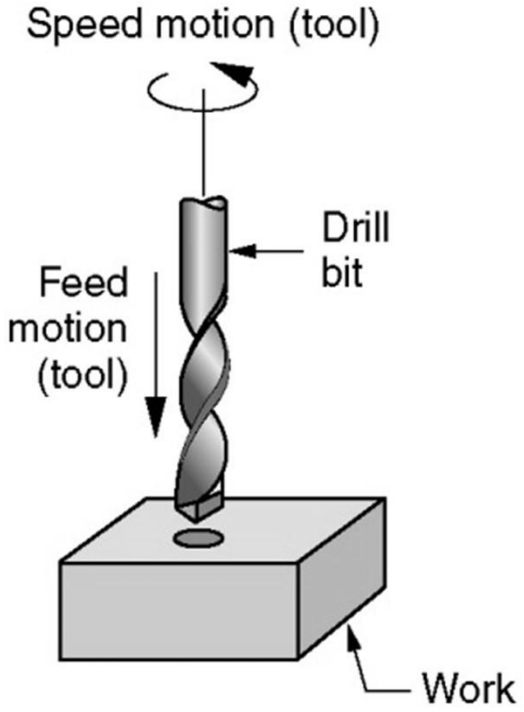
- One drawback of machining: removed material is wasted
- The most common machining methods are *drilling*, *sawing*, *milling*, and *turning*.

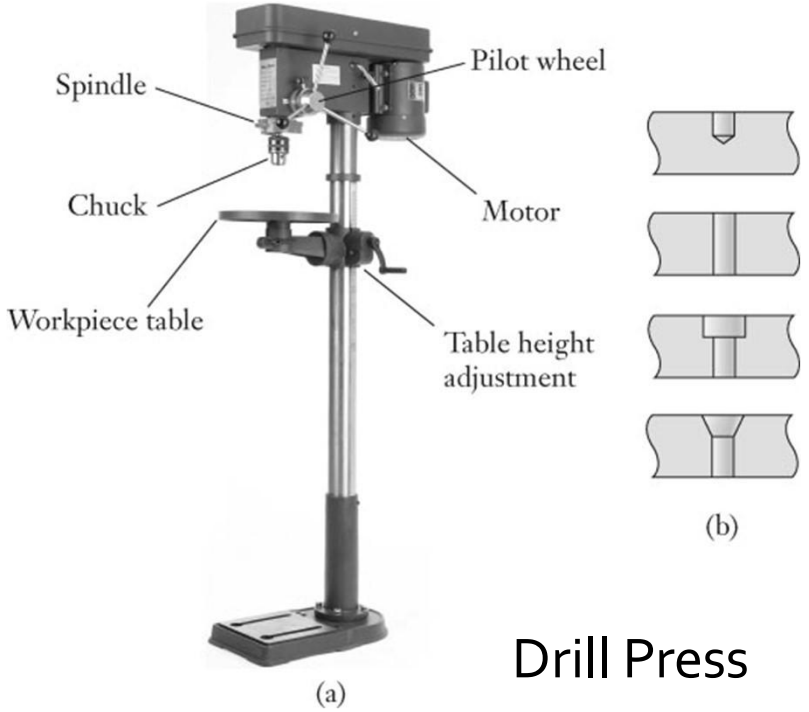


Hydraulic valve assembly
Left: cast from aluminum
Right: machined in order to produce holes, flatten surfaces, and cut threads

Manufacturing Processes: Machining - Drilling

- Drilling is a cutting process that uses a drill bit to cut a hole of circular cross-section in solid materials. The drill bit is usually a rotary cutting tool, often multipoint.
- A drill bit is held in the rotating chuck, and as a machinist turns the pilot wheel, the bit is lowered into the workpiece's surface.



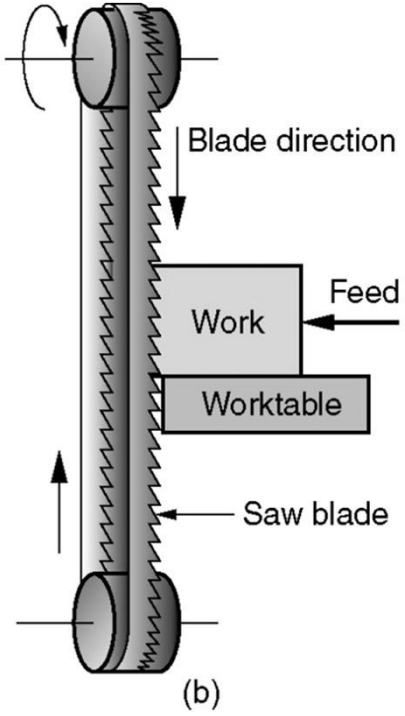


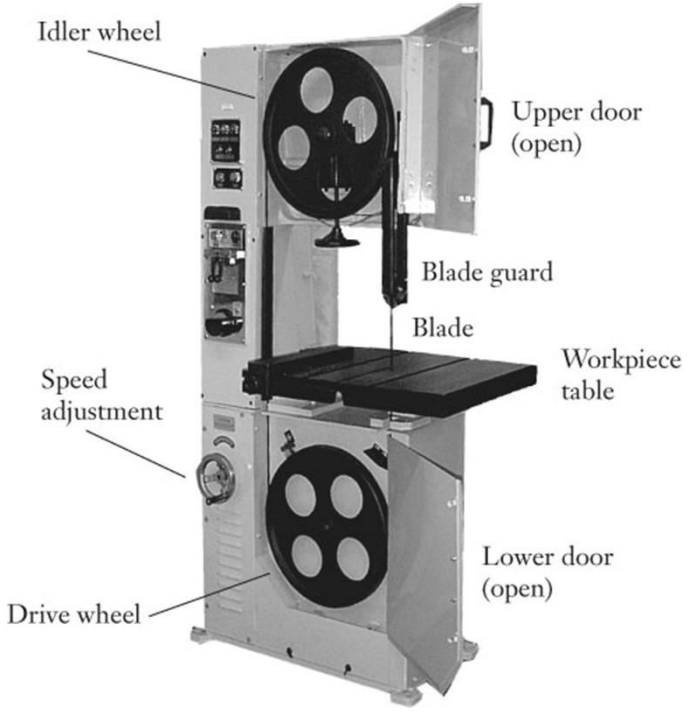
Drill Press

Hole making

Manufacturing Processes: Machining - Sawing

- In sawing a machinist will use a band saw to make rough cuts through metal.
- The blade is a long, continuous loop that has sharp teeth on one edge, and it rides on the drive and idler wheels.





Band Saw

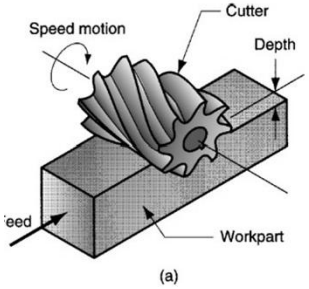
Manufacturing Processes: Machining - Milling

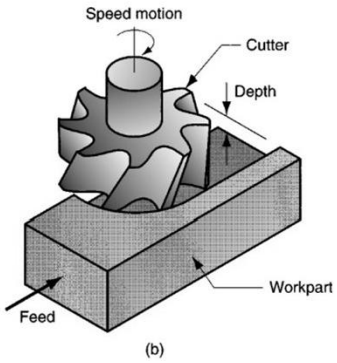
- A milling machine (or mill) is used for machining the rough surfaces of a workpiece flat and smooth, and for cutting slots, grooves, and holes.
- The milling machine is a versatile machine tool in which the workpiece is moved slowly relative to a rotating cutting tool.

Milling Machine





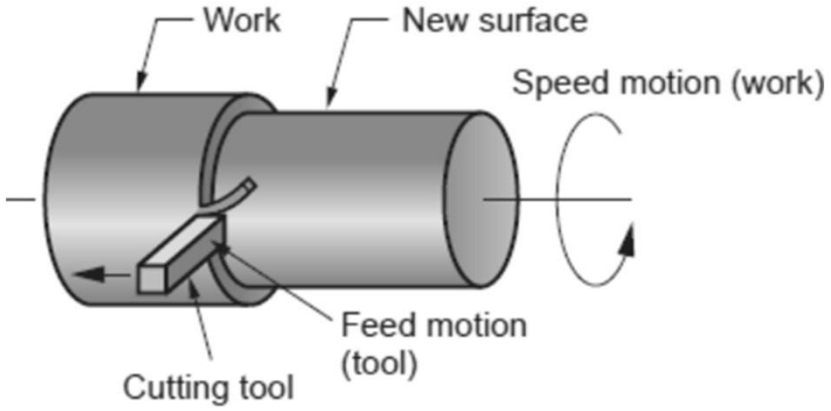


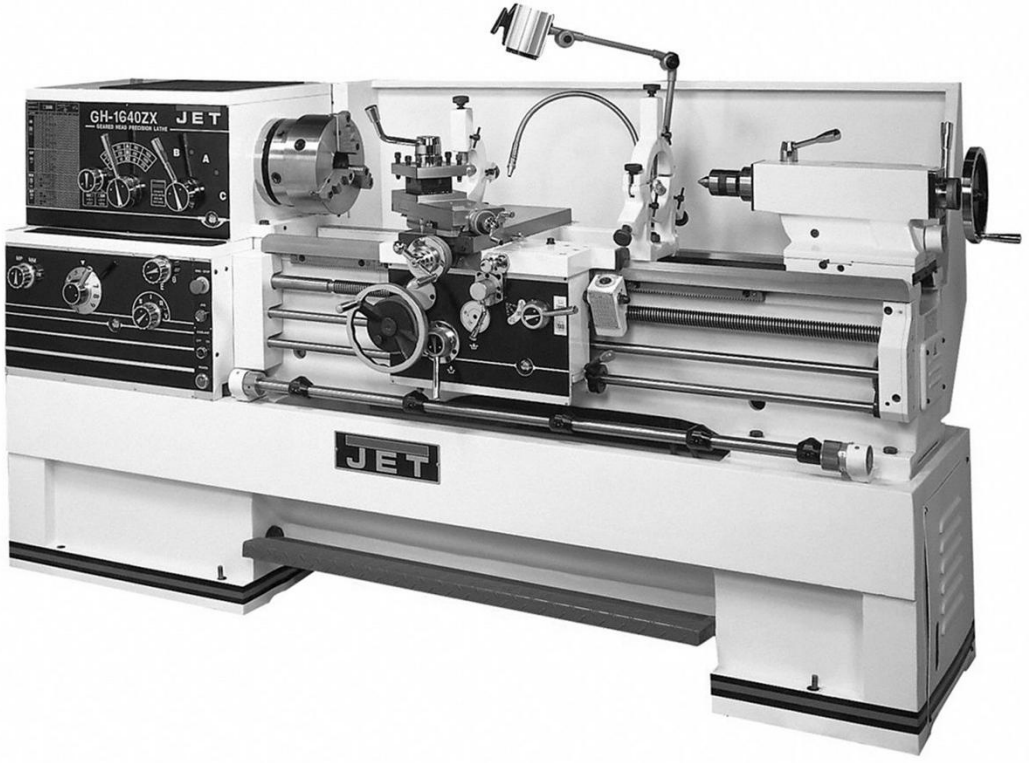


Milling Process

Manufacturing Processes: Machining - Turning

- In turning a lathe holds a workpiece and rotates it about the centerline as a sharpened tool removes chips of material.
- The lathe is used to produce cylindrical shapes and other axisymmetrical components.



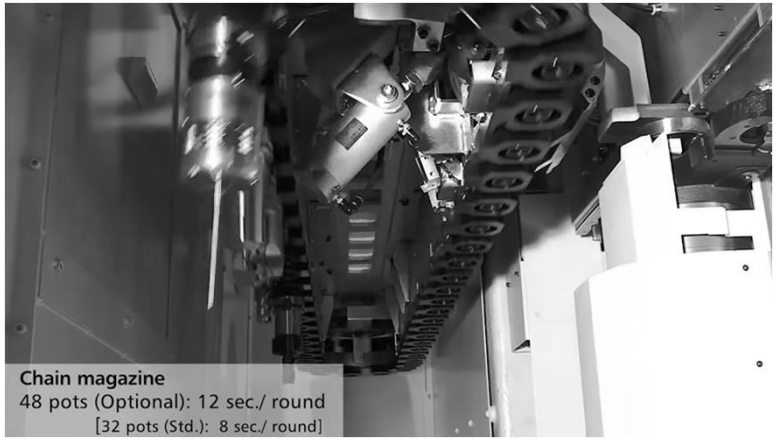


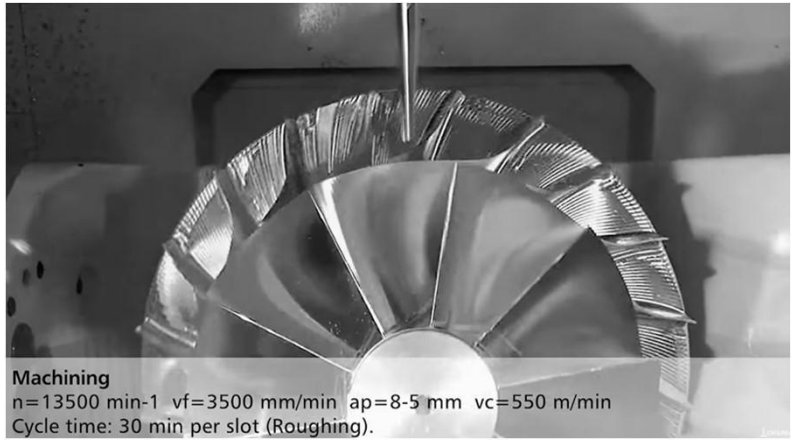
Manufacturing Tools: Numerical Control

- Computer-aided manufacturing (CAM) uses computers to control machine tools to cut and shape materials.
- Generally used when high precision is required, or when a repetitive task must be performed on a large scale.
- Computer-numerically-controlled (CNC) machine tools allow production of parts from a computer-generated drawing.



CNC machining center





Manufacturing Processes: Joining and Finishing



- Joining operations are used to assemble subcomponents into a final product by *welding*, *soldering*, *riveting*, *bolting*, or *adhesively bonding* them.
- Many bicycle frames, for instance, are welded together from individual pieces of metal tubing.

How A Colnago Steel Frame Is Made

vs

Advanced Robotic Bike Frame Welding

- Finishing steps are taken to treat a component's surface in order to make it harder, improve its appearance, or protect it from the environment.
- Some finishing processes include *polishing*, *electroplating*, *anodizing*, and *painting*.

Friction Welding

Additive Manufacturing

- 1980s
- Fabricating artifacts in layers via different technologies
- First called Rapid Prototyping – now not only for prototyping
- 3D printing

The advantages of additive manufacturing:

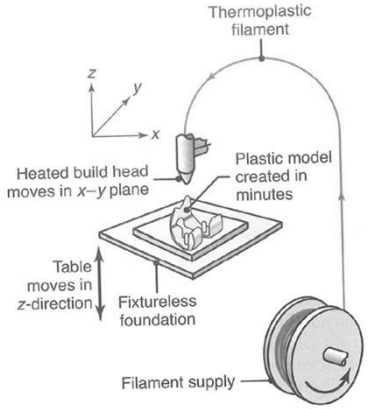
- The cost of entry is low.
- Savings on waste material and energy.
- Low prototyping cost.
- Most suitable for small volume manufacturing.
- You can manufacture assemblies as a single part.
- It support lattice structure manufacturing.
- Customizations to fulfil customer demands is possible.

Additive Manufacturing

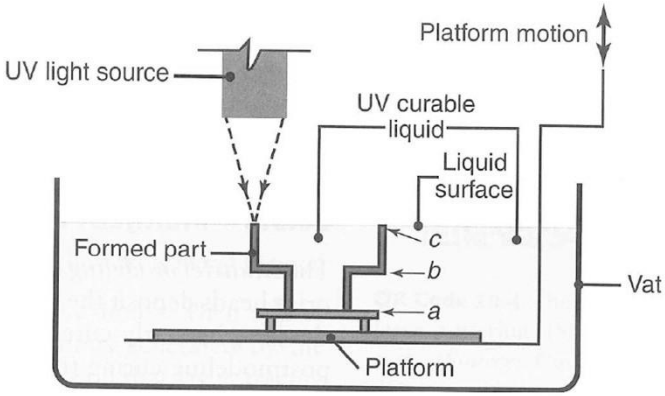




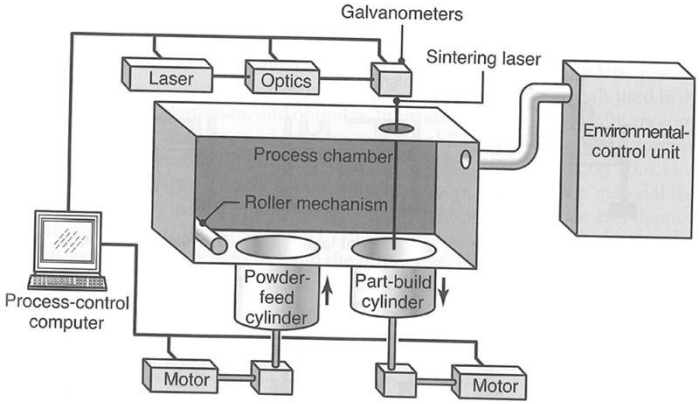
Fused deposition modeling (FDM)



Stereolithography



Selective laser sintering





Ultimaker 3



Formlabs Form 3



Formlabs Fuse 1

5. WEEK

Effective Writing and Presentation Skills

ME 110 Introduction to Mechanical Engineering

Lecture Slides #4

Unit Systems and Conversions

- A unit is an arbitrary division of a physical quantity, which has a magnitude that is agreed upon by mutual consent.
- There are two main unit systems used in engineering:
  - United States Customary System (USCS)
  - International System of Units (SI)
- Both the USCS and SI are made up of base units and derived units.

Base and Derived Units

- Base units are fundamental quantities that cannot be expressed in terms of any simpler elements.
- Derived units are combinations of several different base units.
- For example,
  - Length is a primary dimension, whose base unit is meter (m) in the SI and foot (ft) in the USCS.
  - Velocity is a secondary dimension (length/time), whose derived unit is (m/s) in the SI and (ft/s) in the USCS.

Base Units

Table 3.1

Base Units in the SI

Quantity	SI Base Unit	Abbreviation
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	Kelvin	K
Amount of substance	mole	mol
Light intensity	candela	cd

Table 3.4

Base Units in the USCS

Quantity	USCS Base Unit	Abbreviation
Length	foot	ft
Force	pound	lb
Time	second	s
Electric current	ampere	A
Temperature	degree Rankine	°R
Amount of substance	mole	mol
Light intensity	candela	cd

Common Derived Units (SI)

Table 3.2

Certain Derived Units in the SI

Quantity	SI Derived Unit	Abbreviation	Definition
Length	micrometer or micron	μm	1 μm = 10^{-6} m
Volume	liter	L	1 L = 0.001 m^3
Force	newton	N	1 N = 1 (kg · m)/s^2
Torque, or moment of a force	newton-meter	N · m	—
Pressure or stress	pascal	Pa	1 Pa = 1 N/m^2
Energy, work, or heat	joule	J	1 J = 1 N · m
Power	watt	W	1 W = 1 J/s
Temperature	degree Celsius	°C	°C = K − 273.15
Although a change in temperature of 1 Kelvin equals a change of 1 degree Celsius, numerical values are converted using the formula.			

Common Derived Units (USCS)

Table 3.5

Certain Derived Units in the USCS

Quantity	Derived Unit	Abbreviation	Definition
Length	mil	mil	1 mil = 0.001 in.
	inch	in.	1 in. = 0.0833 ft
	mile	mi	1 mi = 5280 ft
Volume	gallon	gal	1 gal = 0.1337 ft^3
Mass	slug	slug	1 slug = 1 $(lb \cdot s^2)/ft$
	pound-mass	lbm	1 lbm = 3.1081×10^{-2} $(lb \cdot s^2)/ft$
Force	ounce	oz	1 oz = 0.0625 lb
	ton	ton	1 ton = 2000 lb
Torque, or moment of a force	foot-pound	ft · lb	—
Pressure or stress	pound/$inch^2$	psi	1 psi = 1 lb/in^2
Energy, work, or heat	foot-pound	ft · lb	—
	British thermal unit	Btu	1 Btu = 778.2 ft · lb
Power	horsepower	hp	1 hp = 550 $(ft \cdot lb)/s$
Temperature	degree Fahrenheit	°F	°F = °R – 459.67

Although a change in temperature of 1° Rankine also equals a change of 1° F, numerical values are converted using the formula.

Converting Between the SI and USCS

- Unit conversion factors enable engineers to convert values from USCS to an equivalent value in SI and vice versa.
- The conversion process involves changes being made both to the numerical value and to the units associated with it.
- See the book and other resources for conversion of units. Most textbooks have unit conversion tables inside their covers.

Mars Climate Orbiter


MARS
CLIMATE
ORBITER
-$300M

Conversion Factors

Mass and Density

1 kg = 2.2046 lb
1 g/cm^3 = 10^3 kg/m^3
~~1 g/cm^3 = 62.428 lb/ft^3~~
1 lb = 0.4536 kg
1 lb/ft^3 = 0.016018 g/cm^3
1 lb/ft^3 = 16.018 kg/m^3

Length

1 cm = 0.3937 in.
1 m = 3.2808 ft
1 in. = 2.54 cm
1 ft = 0.3048 m

Velocity

1 km/h = 0.62137 mile/h
1 mile/h = 1.6093 km/h

Volume

1 cm^3 = 0.061024 in.^{3}
1 m^3 = 35.315 ft^3
1 L = 10^{-3} m^3
1 L = 0.0353 ft^3
1 in.^{3} = 16.387 cm^3
1 ft^3 = 0.028317 m^3
1 gal = 0.13368 ft^3
1 gal = 3.7854×10^{-3} m^3

Force

1 N = 1 kg · m/s^2
1 N = 0.22481 lbf
1 lbf = 32.174 lb · ft/s^2
1 lbf = 4.4482 N

Pressure

1 Pa = 1 N/m^2
= 1.4504×10^{-4} lbf/in.^{2}
1 bar = 10^5 N/m^2
1 atm = 1.01325 bar
1 lbf/in.^{2} = 6894.8 Pa
1 lbf/in.^{2} = 144 lbf/ft^2
1 atm = 14.696 lbf/in.^{2}

Energy and Specific Energy

1 J = 1 N · m = 0.73756 ft · lbf
1 kJ = 737.56 ft · lbf
1 kJ = 0.9478 Btu
1 kJ/kg = 0.42992 Btu/lb
1 ft · lbf = 1.35582 J
1 Btu = 778.17 ft · lbf
1 Btu = 1.0551 kJ
1 Btu/lb = 2.326 kJ/kg
1 kcal = 4.1868 kJ

Energy Transfer Rate

1 W = 1 J/s = 3.413 Btu/h
1 kW = 1.341 hp
1 Btu/h = 0.293 W
1 hp = 2545 Btu/h
1 hp = 550 ft · lbf/s
1 hp = 0.7457 kW

Specific Heat

1 kJ/kg · K = 0.238846 Btu/lb · °R
1 kcal/kg · K = 1 Btu/lb · °R
1 Btu/h · °R = 4.1868 kJ/kg · K

Others

1 ton of refrigeration = 200 Btu/min = 211 kJ/min
1 volt = 1 watt per ampere

From M.J. Moran, H.N. Shapiro, D.D. Boettner, M.B.Bailey, Principles of Engineering Thermodynamics, SI version, 8th ed., Wiley, 2015.

Dimensional Consistency

- Dimensional consistency means that the units associated with the numerical values on each side of an equality sign match.
- It is a great way to double-check your calculation as well.

Exercise: Dimensional Consistency

A mercury-in-glass barometer reads 750 mm Hg. Find the atmospheric pressure in Pascals.

$\rho_{Hg} = 13.59\ \text{g/cm}^3$, $g = 9.81\ \text{m/s}^2$

$p_{atm} = \rho_{Hg} g H = (13.59\ \text{g/cm}^3)(9.81\ \text{m/s}^2)(750\ \text{mm})$

$= 99988$???

Carefully convert units:

$$= \left[(13.59)\left(\frac{\text{g}}{\text{cm}^3}\right)\left(\frac{1}{1000}\frac{\text{kg}}{\text{g}}\right)\left(100\frac{\text{cm}}{\text{m}}\right)^3\right]\left[9.81\frac{\text{m}}{\text{s}^2}\right]\left[(750)(\text{mm})\left(\frac{1}{1000}\frac{\text{m}}{\text{mm}}\right)\right]\left[\frac{1\ \text{N}}{1\left(\text{kg}\cdot\frac{\text{m}}{\text{s}^2}\right)}\right]$$

$$= 99988\frac{\text{N}}{\text{m}^2} \cong 10^5\ \text{Pa}$$

Order of Magnitude Prefixes

Table 3.3

Order-of-Magnitude Prefixes in the SI

Name	Symbol	Multiplicative Factor
tera	T	$1{,}000{,}000{,}000{,}000 = 10^{12}$
giga	G	$1{,}000{,}000{,}000 = 10^{9}$
mega	M	$1{,}000{,}000 = 10^{6}$
kilo	k	$1000 = 10^{3}$
hecto	h	$100 = 10^{2}$
deca	da	$10 = 10^{1}$
deci	d	$0.1 = 10^{-1}$
centi	c	$0.01 = 10^{-2}$
milli	m	$0.001 = 10^{-3}$
micro	μ	$0.000{,}001 = 10^{-6}$
nano	n	$0.000{,}000{,}001 = 10^{-9}$
pico	p	$0.000{,}000{,}000{,}001 = 10^{-12}$

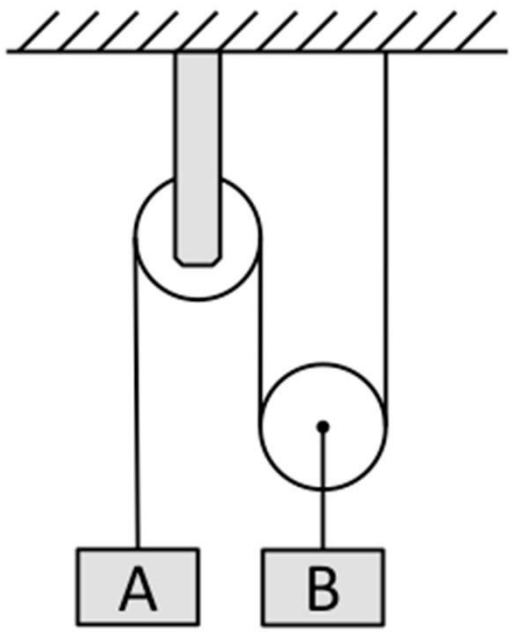
Exercises: Common Quantities

1. The max/min pressures for human blood [mmHg, kPa]
   - 120 and 80 mmHg
   - 16 and 11 kPa
2. Standard automobile/truck tire pressure [kPa, psi]
   - 30 and 90 psi
   - 2 and 6 bar
3. Diameter and temperature of the sun [km, K]
   - 695 700 km
   - 5 778 K (not ^{0}K)
4. Lightbulb [W, hp] (LED)
   - 10 w
   - 0.013 hp
5. Density of air, water [kg/m^3]
   - 1 kg/m^3
   - 1000 kg/m^3
6. Standard automobile torque [N·m]
   - 100 N·m

Assumptions and Approximations

- In the earlier stages of design and analysis, engineers nearly always make assumptions and approximations when they solve technical problems.
- Assumptions are made to reduce a real system, as imperfect and nonideal as it may be, into its most basic and essential elements; a workable model.
- Approximations are made to remove extraneous factors that complicate the problem, especially numerically, but that otherwise have little influence on a final result.
- The solutions obtained through many assumptions and approximations will generally be accurate to perhaps a factor of 10. Thus, the answer will be accurate within its order-of-magnitude.

Pulley - Mass System



Possible Assumptions

- Neglect the masses of the pulleys
- Neglect friction in the pulley bearings
- Neglect the masses of ropes
- The ropes do not stretch
- The ropes do not slide on the pulleys
- Take masses A and B as point masses

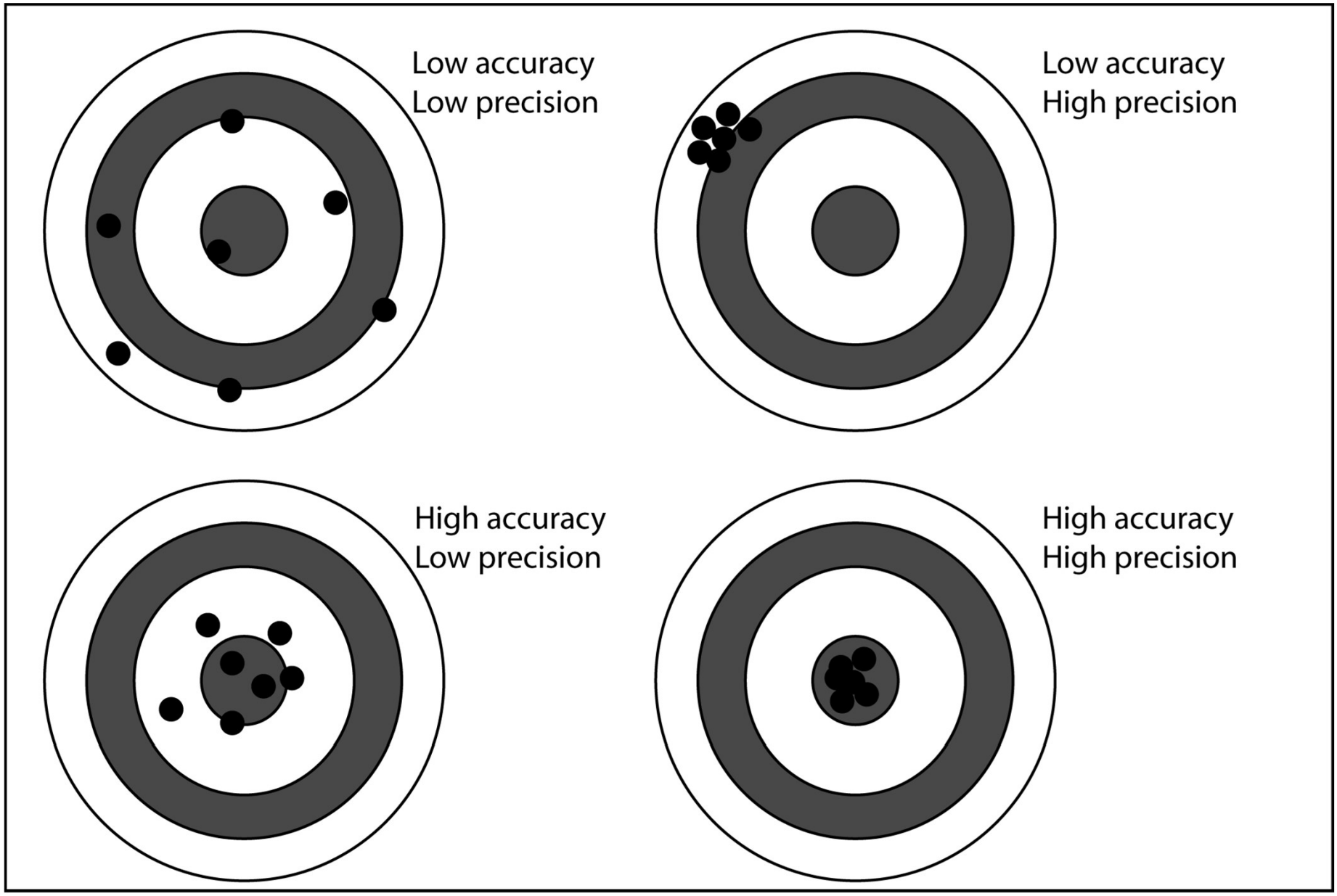
Possible Approximation

- Take $g = 9.8\ m/s^2$

Significant Figures

- All measurements have some degree of uncertainty associated with them. To reflect this, we state a numerical result with an appropriate number of significant figures/digits.
- The number of significant digits in an answer to a calculation will depend on the number of significant digits in the given data.
- A significant digit is one that is known to be precise despite the inaccuracy present in the supplied information.
- Accuracy and precision

Accuracy and Precision



Rules for Significant Figures

- The left-most non-zero digit is the first significant figure and is called the most significant figure.
- Least significant figure:
  - In a number with a decimal point present, the right-most digit (zero or non-zero) is the least significant figure.
  - In a number without a decimal point present, the right-most non-zero digit is the usually least significant figure.
- All digits between the most significant figure and the least significant figure are significant.
- When a number ends in zeroes that are not to the right of a decimal point, the zeroes are not necessarily significant. In this particular case, the number of significant figures should be explicitly indicated somewhere. This ambiguity can be eliminated if the number is expressed in scientific notation.

$m \times 10^n$, $1 \leq m < 10$, n is any integer

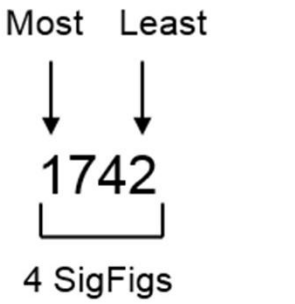
Scientific Notation

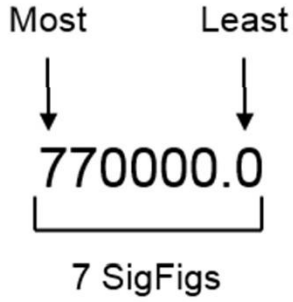
Guesstimation and Scientific Notation / Enrico Fermi

Exercise: Significant Figures

Determine the # of significant figures in the following numbers

1742	55.00003	0.000668	229006
770000.0	0.00097000	280000	2.800×10^5

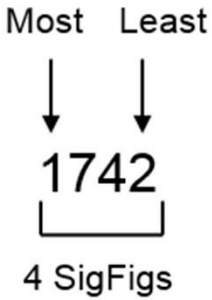


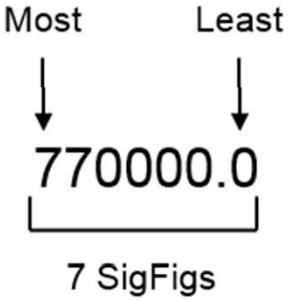


Exercise: Significant Figures

Determine the # of significant figures in the following numbers

1742	55.00003	0.000668	229006
770000.0	0.00097000	280000	2.800×10^5





55.00003 →7 Sig Figs
(Zeros between non-zero digits are significant.)

0.00097000 →5 Sig Figs
(Leading zeros are not significant; 9, 7, and the three trailing zeros are significant.)

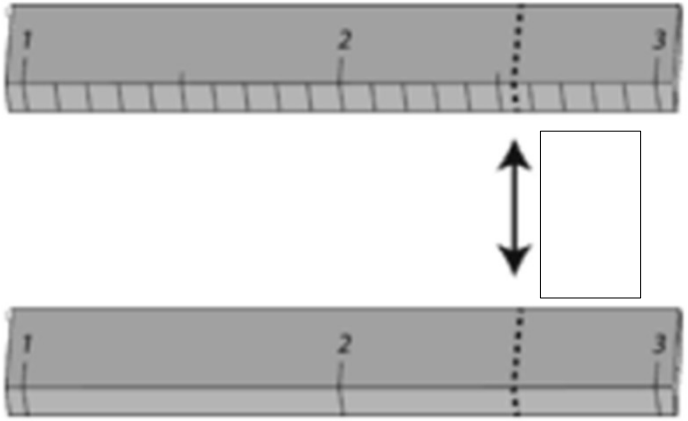
0.000668 →3 Sig Figs
(Leading zeros merely indicate decimal placement and are not significant; only 6, 6, and 8 count.)

28000 →2 Sig Figs
(Trailing zeros in a whole number without an explicit decimal point are considered ambiguous/insignificant.)

229006 →6 Sig Figs
(Zeros trapped between non-zero digits are significant.)

2.800×10^5 →4 Sig Figs
(In scientific notation, the exponential part is ignored; 2, 8, and both trailing zeros after the decimal are significant.)

Reporting Significant Figures in Measurements



- Number of significant figures: precision of the measurement device + 1 estimated digit

Source

Rounding Rules

- In computing intermediate steps, carry as many digits, significant or not, as possible/reasonable.
- Once all intermediate calculations are conducted, the final result must be rounded to the appropriate number of significant figures.
- See Significant Figures documents on classroom and details of Rounding

In-Class exercise

Report Writing

- The report is a common method of communication used by engineers to convey their ideas and solutions to other engineers or their managers.
- It is critical to put your ideas into proper context, but you must select your audience. You should organize and design your data to meet the audience's technical knowledge and information needs.

General Report Sections

- Title Page – Only basic information identifying the report such as names, affiliations, date, etc.
- Abstract – An accurate statement of the subject of the report and the nature of the obtained results.
  - It is written last
  - It provides enough information to decide whether the entire report is of interest to the reader
  - Do not repeat the title in the abstract
  - Do not report extensive numerical results in the abstract
  - Do not use footnotes, references, acknowledgements, acronyms or unfamiliar abbreviations
  - Limit length to about 200 words

General Report Sections

- Introduction – Provides the precise subject of the work and explicitly states all objectives. References are typically cited extensively in the introduction.
- Theory – Provides the reader details of the proposed theory governing the phenomenon of interest.
- Data and Results – Presents the reader “reduced data” or raw data from the experiments or other work done, that has been analyzed and prepared for interpretation, in the form of tables, figures, etc.
- Discussion – Data and results are interpreted and discussed.

General Report Sections

- Conclusion – Reviews the principal results of the report and includes an honest discussion of the limitations, if any, of these results.
- References – The sources of referenced work, such as textbooks, journal articles, manufacturer literature, etc.
- Appendices – Generally, raw data, sample calculations and extensive diagrams are placed here.

GRADING RUBRIC

PROBLEM 1 (TOTAL POINTS: 10)



WWW.PHDCOMICS.COM

Common Issues

- Some highlights:
  - All writing must be formal.
  - Use a common, non-decorative font (such as Arial or Times New Roman) with a font size of 11/12 pt
  - All pages are numbered with Arabic numerals starting with '1' on the first page after the title page.
  - Writing throughout the text must be in the present tense.

    "… is discussed further below" not "… will be discussed further later"

Common Issues

- Do not use any personal pronouns.

  "… it was determined that…" not "… we determined that…"
- All acronyms/abbreviations, except those found in standard dictionaries, are to be defined the first time they appear.
- Do not use any gratuitous terms such as obviously, well known, easily, or trivial. This tends to insult the reader if the concept is not immediately obvious or known.
- A report is not improved by increasing its page count. A short, accurate, and concise report is far better than one that is "bulked up" with needless information.
- The word data is inherently plural and therefore any verb referring to it must be in the plural form.

  "The data are obtained using …" not "The data is obtained using …"

  "The data show that …" not "The data shows that …"

Tables

Incorrect

Table 4 Characteristics of a Piezo-electric Accelerometer Output Experiencing Sinusoidal Acceleration*

Peak Acceleration	Peak E
0.521 m/s^2	0.109 V
0.899	0.156 V
2.43	0.277 V
5.68	1.46 V
12.5	7.94 V
25.3	16.7 V
67.3	33.4 V

*Note that the first point was obtained using different instrument input range than the others.

Correct

Table IV Piezo-electric Accelerometer Output Experiencing Sinusoidal Acceleration

Peak acceleration (m/s^2)	Peak voltage (V)
0.521	0.109[a]
0.899	0.156
2.43	0.277
5.68	1.46
12.5	7.94
25.3	16.7
67.3	33.4

[a] This point was obtained using a different instrument input range than the others.

- A descriptive caption, not a title, is placed immediately above a table.
- Do not use too many lines to create boxes or columns
- Footnotes are designated by superscripted lowercase letters or characters.
- Units should be indicated on headings and not on each of the table values.

Figures

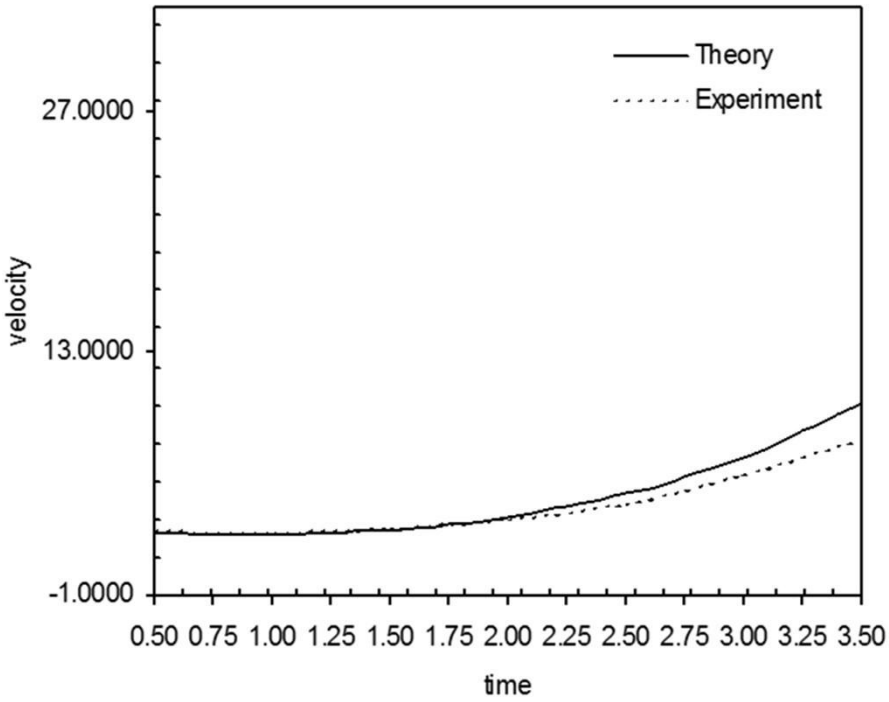


figure III: time vs velocity

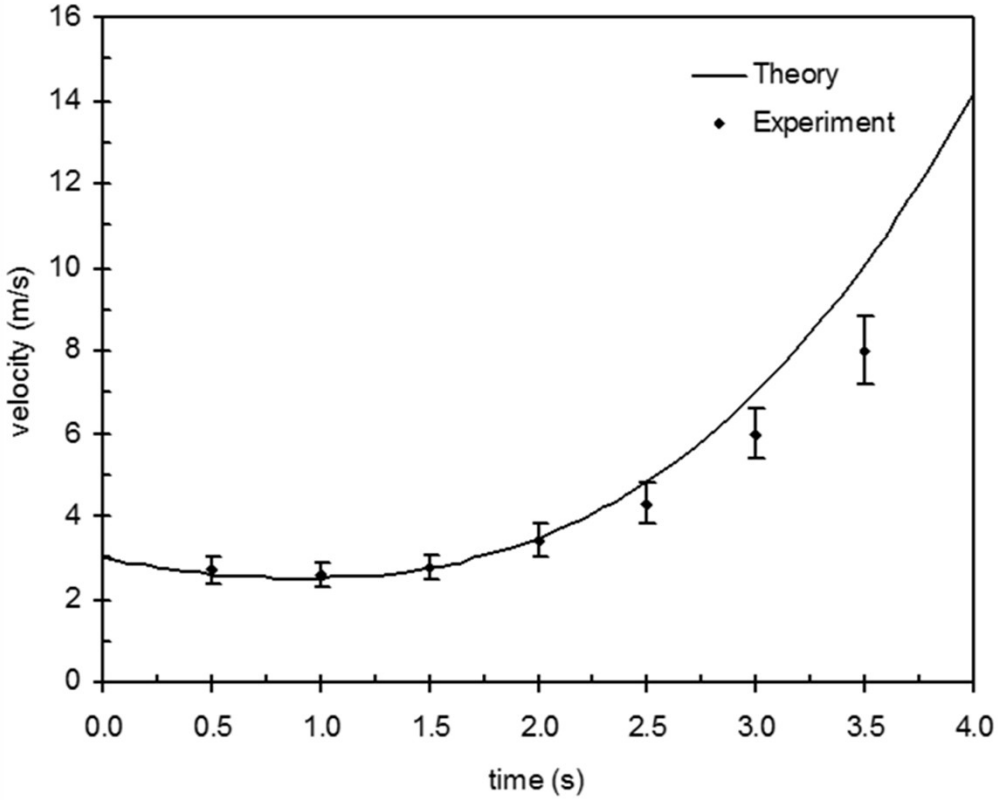


Figure 3. Comparison of theoretically predicted vehicle velocity (post boost stage) to values experimentally determined by vehicle-mounted accelerometers. Error bars correspond to a 95% CI of the mean.

- Both figure titles and figure captions are commonly used
- Do not restate the axis of the figure
- Select proper number of significant digits, axis labels, major/minor gridlines.
- Use units
- Use error bars if necessary

Presentations (Poster or Powerpoint)

- Presentations allow you to convey your knowledge and findings to a live audience in a limited time.
- The technical content and format of the presentation, as well as your performance is important.
- Prepare your presentation for the audience: Supervisor? Coworker? Non-engineer? International?
- Think about what information your audience already has, so that you will not go into unnecessary detail, or avoid giving important information (a delicate balance).
- Perform a spelling and grammar check.
- Practice, so that you will not go overtime.

Presentations (Poster or Powerpoint)

- Write bulleted points instead of complete sentences and keep it simple.
- Some dressing up is generally expected, but don't overdo!
- Make eye contact with audience members.
- Be audible, but don't shout.
- If it's a team presentation, each team member should be active during the presentation.
- If multiple presentations are taking place in a session, you are expected to show full attention and respect to others.



Presentations (Powerpoint)

- The general rule is 1 minute per slide and six lines per slide.
- Do not make your slides crowded, keep them simple
- Use a font size of at least 24 pt., use a font type that is sans serif (e.g. Arial is sans serif, while Times New Roman is serif and harder to read on a presentation).

WARNING



Presentations that are shared in class and also distributed as lecture notes as reading material may not conform to above guidelines.

Emails – Bad Examples

Me110 Projesiyle ilgili bir sorum olacak. Eğer biz arabamızda helium balonu kullanırsak arabamızın ağırlığını 0 veya 0a yakın yapabiliriz. Böylece yol/kilo sonsuz olur.

Merhaba [ ],

Soru sormayı unutmuşsun.

AGY

****

Hocam iki seminere de dersim çakışıyor ne yapmalıyım?

Hocam merhaba. Dersinizi almak icin dilekçe verdim fakat henüz odtuclass da ders gözükmüyor bu yüzden ödev de gözükmüyor.yapmam gereken bir durum var mı yoksa ödevi yapmaya devam etmeli miyim? İyi çalışmalar

Merhaba [ ],

Hangi ders? Biraz daha iyi e-posta becerileri lütfen!

AGY

****



Emails

- Be sure to correctly address the recipient. If you are unsure of a person's proper title (Dr., Mrs.), look it up!
- Use an appropriate subject line. Avoid silly subjects (Mutlaka okumalısın!) or omitting the subject line—this may cause the e-mail to end up in the Junk Mail folder.
- Sign your full name and include contact information if necessary. When sending e-mail about a class, include your course number.
- Change your sending name to your full name (such as Ahmet Yılmaz) and do not use funny nicknames (No alemlerinkrali@gmail.com)

Emails

- Keep it brief. Do not use one continuous paragraph.
- If you expect a response, be sure that your request is clear.
- Use correct capitalization and punctuation.
- Avoid IM talk, like LOL, KIB
- Avoid putting anything in e-mail you would not say in person.
- To avoid sending an e-mail before you have a chance to check over your work, fill in the To: and CC: lines last. Hint: Gmail has an UNDO option!

Emails – Good

İyi akşamlar hocam,

Daha önceden, akademik takvime göre yarıyıl tatiline denk geldiği için 21 – 24 Ocak 2018 tarihleri arasında düzenlenen CERN gezisine kayıt yaptırmıştım ancak 8/12/2017 tarihinde odtuclass'a yüklediğiniz ME 110 grup projesinin sunum tarihiyle kesiştiğini gördüm. Eğer bir sorun teşkil etmeyecekse verdiğiniz ödevi önceden sunmamız veya sömestrdan önce bitireceğimiz projeyi grup arkadaşlarımın sunması gibi bir imkanımız olur mu? Gerektiği takdirde geziye kayıt ve davet belgelerimi size gönderebilirim.

Saygılarımla ,

22

Dear Madam,

My name is (210) and I am a second year,undergraduate student from , studying in mechanical engineering department. I took Thermodynamics 1 Course (ME 203) from you in 2015-2016 Fall semester.

Earlier this month, I told you in person that I am in the process of doing my internship applications via the website called http://www The website has a section for academic references. The section requires me to attach a PDF document of my reference letter which will go along my online resume.

Since I was your student for a very major course,I would be really grateful if you could give me an academic reference letter. As I have to upload the letter online, I will need a PDF document of the letter which you can send via email.

I hope my query is sufficiently clear to you. Looking forward to your cooperation.

Regards,


İLKOKUL MEZUNU BİLE DEĞİLSİNİZ AMA GOOGLE'DA MÜDÜR OLDUNUZ...BU İNANILMAZ BİR BAŞARI HİKAYESİ...NASIL OLDU BU?.. BİRAZ KENDİNİZDEN BAHSEDER MİSİNİZ ?..
BABAM ÇİFTÇİ ESNAFLIK VE ANNESİ EV HANIMI OLAN VE ÇİFTLİKLE BABA YARDIMCI OLUR... BİZ ÜÇ KARDEŞİZ VE BEN YAŞLI BİRİYİM ...BEN ÖNCEDEN TECRÜBEM BİR YIL VAR...GOOGLE TRANSLATE MÜDÜRÜ BENİ SEÇİLDİ...
ULAN...SEN MİSİN BUNCA ZAMANDIR MİLLETİN ANASINI AĞLATAN?..
BEN OLUR...
DOĞRU DÜZGÜN KONUŞSANA LAN!..
MAL BİRAZ BEN.
UYKUSUZ